Student Management System (Full-Stack)

PHP | MySQL | Bootstrap

D.Meenakshi

Introduction

- A **full-stack web application** developed to manage student data efficiently.
- Supports both **Admin and Student roles** with role-based access control.
- Provides features like attendance tracking, marks management, and report generation.
- Includes secure functionalities such as OTP-based password reset and document storage.
- Helps in reducing manual work and improving data accuracy.

Objectives

- **Automate Student Management**
  Replace manual processes with a digital system for efficiency.
- **Role-Based Access Control**
  Provide separate access for Admin and Students with different permissions.
- **Ensure Data Security**
  Protect user data using authentication, OTP verification, and secure storage.
- **Improve Data Accuracy**
  Reduce human errors in attendance, marks, and records.
- **Enhance User Experience**
  Provide a simple, responsive, and easy-to-use interface.

Tech Stack

Purpose of Each Technology

- **PHP**
  Used to handle backend logic such as login authentication, CRUD operations (Add, Update, Delete students), and processing attendance & marks.
- **MySQL**
  Stores all project data including student details, attendance records, marks, login credentials, and uploaded documents.
- **Bootstrap 5**
  Used to design a responsive and user-friendly interface for Admin and Student dashboards.
- **JavaScript**
  Handles client-side validation, dynamic interactions, and improves user experience.
- **PHPMailer**
  Used to send OTP emails for password reset functionality securely.
- **Composer**
  Manages external libraries like PHPMailer and ensures proper dependency handling

System Architecture

System Flow

Login

Authenticate

Access Dashboard

Manage Data

Logout

Admin Dashboard

- **Manage Students**
  Add, update, and delete student records efficiently.
- **View Reports**
  Access detailed reports including attendance, marks, and performance.
- **Update Academic Data**
  Enter and modify semester marks, CGPA, and attendance records.
- **Monitor System Activity**
  Track overall system usage and student data updates.
- **Generate Reports**
  Create downloadable PDF reports for academic records.

Student Dashboard

- **View Profile**
Students can view and manage their personal information.
- **Check Marks & CGPA**
Access semester-wise marks and overall CGPA.
- **Download Reports**
Generate and download academic reports in PDF format.
- **View Attendance**
Monitor attendance records subject-wise.
- **Access Document Wallet**
Upload and retrieve important documents securely.

Project Deployment

How the Project is Deployed

- The project is deployed locally using a web server environment (XAMPP).
- PHP files are executed through Apache server.
- MySQL database is connected using phpMyAdmin.

Where the Project is Deployed

- Deployed on **Localhost (Local Server)**

Step-by-Step Deployment Process

- Download/Clone the project from GitHub
- Install XAMPP server
- Copy project folder into htdocs directory
- Start **Apache** and **MySQL** in XAMPP
- Open **phpMyAdmin**
- Create a new database
- Import the provided .sql file
- Configure database connection in PHP files

Project Repository Link

GitHub Link:

https://github.com/dakameeenakshi-lab/Student-Management-System.git